

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 37

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm²) • They use the polyp's CO₂ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently • They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients • Polyps also trap plankton with cilia and stinging nematocysts; surplus CO₂ is used to build the calcium carbonate skeleton	4
01.2	• High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water • Found only where all these coincide (about 1% of ocean area) • Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance	3
01.3	<i>Any 3 from:</i> • Fisheries: support over a quarter of marine species including fish, lobsters, conch • Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores • Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts • Climate control: carbonate skeletons sequester CO₂ • Tourism (diving, snorkelling)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	• As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) • As otters recovered after 1990 urchins declined and kelp recovered • Otter and kelp trends are positively correlated; urchins negatively correlated with both	3
02.2	• Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise • Sea urchins graze kelp so the kelp forest was over-grazed and destroyed • Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance	3
02.3	<i>Any 2 from:</i> • Over-fishing of sandeels reduced puffins • Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish • Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees	2
02.4	• They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open • Species that depend on these features (light, water, minerals) cannot survive in their absence	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
03.1	• Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned • Loss of canopy and litter exposes soil to rain splash and wind • Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) • Reduced organic matter and biota lower infiltration and water retention	4
03.2	• Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers • Increased albedo of cleared land (reflects more sunlight) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change	4
03.3	<i>Any 2 from:</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines • Loss of genetic resources / potential medicines before they are discovered	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
04.1	• The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean • The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly	2

Question	Marking guidance	Marks
04.2	- t-test - Comparing two means of measured (continuous) data	2
04.3	- There is a 30% probability that the difference occurred by random chance - Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted	2
04.4	- Two correlated variables may both be caused by a third variable that was not investigated - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation	2
04.5	- Increase the number of replicates (reduces the effect of variability) - Standardise other variables (soil, water, light) more closely to reduce scatter	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	- B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings - B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects - But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed	3
05.2	Any 3 from: - Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) - Woven plastic cloches / fleece over rows to retain warm air near the ground - Burning oil / gas or using large fans in orchards to prevent frost damage to blossom - Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock	3
05.3	- Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) - Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth	2
05.4	- Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn - Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
06.1	Waste products that are the raw materials for other processes	1
06.2	- Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) - The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2

Question	Marking guidance	Marks
06.3	• Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
07.1	• Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change • No single source could meet demand all year; battery storage on the island grid smooths fluctuations • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated)	3
07.2	<i>Any 3 from:</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection	3
07.3	• Small groups can organise, agree and act quickly; decisions benefit people who see the results directly • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away	2
07.4	• Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
08.1	• Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere • Allowing more plant growth, further stabilising climate and soils	3
08.2	<i>Any 3 from:</i> • Lack of ancient historical / instrumental data • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions • Lack of sophisticated equipment / inability to measure many factors • Limited coordination between researchers; lack of data collection in many areas (oceans, poles)	3

Question	Marking guidance	Marks
08.3	• The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
09.1	• In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching)	2
09.2	<i>Any 4 from:</i> • Is the wild population threatened? • Is there a genetically diverse captive population? • Is in-situ conservation failing? • Is keeping a captive population realistic (habitat size, food, relationships, cost)? • Is release likely to be successful now or in future (suitable habitat, threats removed)?	4
09.3	• Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25